TBXT Hit Identification

Top 4 Picks for Chordoma's Master Regulator

A multi-signal computational pipeline for the G177D variant.

TBXT Hackathon · Pillar VC, Boston · 2026-05-09

Target: TBXT G177D (Brachyury) · PDB `6F59` chain A · Site F (Y88 / D177 / L42)

How we map the 3 judging axes

AXIS 1

Scientific rationale

Target biology · 6-signal pipeline · 7-criterion strict gate

AXIS 2

Compound quality

4 picks · all 100% onepot · all non-covalent · all site F

AXIS 3

Hit ID judgment

Tradeoffs · cross-validation · honest expectations

POOL SCREENED

570

novelty-filtered

STRICT-PASS

137

all 7 criteria

BEST BOLTZ KD

3.2 µM

dual-engine 1.02×

AXIS 1

Scientific rationale

Why these methods, why this target, why these 137 candidates.

The target

AXIS 1

- TBXT G177D (Brachyury) - chordoma's master transcription factor and lineage-defining oncogenic driver
- G177D variant (`rs2305089`) is present in > 90% of Western chordoma cases
- Site F pocket directly engages D177 - the variant residue itself
- G177 is 0% conserved across the 16 T-box paralogs ⇒ structural basis for intrinsic selectivity
- Receptor: PDB `6F59` chain A (G177D + DNA construct - matches CF Labs SPR assay)

> Why site F: TEP-recommended · all best Naar SPR binders predicted at site F · TBXT-unique residues anchor selectivity

The pipeline - 6 orthogonal signals

AXIS 1

Signal	What it catches
Vina ensemble (6 receptor confs)	Geometric fit; receptor flexibility
GNINA CNN pose + pKd	Vina-trap detection; ML affinity
TBXT QSAR (RF + XGBoost on 653 Naar SPR Kd)	Target-specific affinity
Boltz-2 co-folding (two independent runs)	Independent affinity + binder classifier
MMGBSA single-snapshot (top 30)	Free-energy refinement
T-box paralog selectivity (16 paralogs)	Off-target risk

> Each signal has a known failure mode - caught by another signal in the stack

⚠ rewards contact maximization · caught by GNINA + Boltz

⚠ PDB-look bias · caught by QSAR + Boltz

⚠ scaffold OOD on novel chemotypes · caught by docking + Boltz

1.02 - 1.34×.

→ predicted Kd · prob_binder · cross-val ratio

⚠ training-distribution bias · cross-engine controls run-to-run noise

GB-solvation.

→ ΔG (kcal/mol) + decomposition

⚠ single-snapshot bias · short-MD averaging mitigates

reacting.

→ per-paralog binding-score gap

⚠ homology pocket-mapping is approximate · used for ranking, not gating

▼

4 · Consensus aggregation

unweighted mean · top-quartile rerun

Each compound carries a 6-element score vector. Each signal is z-scored across the 570-pool; QSAR / Boltz / MMGBSA are direction-inverted so “more positive = better” everywhere. The consensus is the unweighted mean. Top-quartile compounds are then re-run through Boltz and GNINA at higher exhaustiveness to confirm.

`z(vina) + z(gnina) + z(qsar) + z(boltz) + z(mmgbsa) + z(paralog) → mean → rank` *Unweighted because no target-specific calibration set exists for a learned weighting.*

▼

5 · STRICT GATE (7 CRITERIA)

570 → 137

Hard pass / fail; no soft-penalty trade-offs that could allow disqualification by a fail in any single criterion.

C1
onepot.ai 100% catalog match (muni.bio)

C2
strictly non-covalent (SMARTS — incl. boronic acids)

C3
Chordoma rule: MW≤600 · LogP≤6 · HBD≤6 · HBA≤12

C4
Lead-like: 10-30 HA · HBD+HBA≤11 · <5 rings · ≤2 fused

C5
PAINS-clean + no aldehydes / diazo / imines / long alkyl

C6
Tanimoto < 0.85 to Naar / TEP / prior_art

C7
ESOL log S > -5 (DMSO @ 10 mM + aqueous @ 50 µM)

▼

6 · FOUR-TIER CLASSIFICATION

T1 GOLD
0
Kd ≤ 5 µM AND low chemistry risk AND low supplier risk. **Empty by design** — no compound in the pool simultaneously satisfies all three. Surfaced honestly rather than relaxed.

T2 SILVER
16
Kd ≤ 10 µM AND ESOL-soluble. The cleanest tier — both top-2 picks live here.

T3 BRONZE
89
Kd ≤ 50 µM, borderline aqueous solubility (DMSO @ 10 mM still works). The bulk of the pool. Picks #3 and #4 (chemotype diversity) drawn from here.

T4 RELAXED
32
Kd ≤ 100 µM. Tail tier; available for chemotype-diversity promotion if SILVER and BRONZE cluster around a single scaffold.

▼

7 · OUTPUTS

HACKATHON JUDGES SUBMISSION
4
Strict-first ranked picks (best Boltz Kd 3.2 µM, dual-engine 1.02× cross-val agreement, $875 total cost, 4/4 site F).

EXPERIMENTAL PROGRAM FIRST BATCH
20
Ranks 5-24 of the strict-pass set. SILVER + BRONZE chemotype mix; uses ~21% of the program's 96-compound budget (3 batches × 32).

CROSS-VALIDATION EVIDENCE
5
Independent lines: dual-engine Boltz, 10-seed GNINA pose σ, ADMET re-rank, catalog membership, methodological-variant cross-checks (5 overnight HPC variants).

The 7-criterion strict filter

AXIS 1

- C1 onepot 100% catalog match - C2 strictly non-covalent - C3 Chordoma rule (MW ≤ 600, LogP ≤ 6, HBD ≤ 6, HBA ≤ 12) - C4 lead-like ideal (10–30 HA · HBD+HBA ≤ 11 · < 5 rings · ≤ 2 fused)

- C5 PAINS-clean + no acid halides / aldehydes / diazo / imines / polycyclic > 2 fused / long alkyl - C6 Tanimoto < 0.85 to Naar / TEP / prior_art - C7 ESOL log S > -5

> Of 570 pool compounds → 137 pass all 7 criteria → 4 picks (ranks 1, 2, 11, 22)

Tier breakdown: T1 GOLD: 0 (empty by design) | T2 SILVER: 16 | T3 BRONZE: 89 | T4 RELAXED: 32

AXIS 2

Compound quality

The 4 picks: properties, evidence, and chemistry.

Picks at a glance

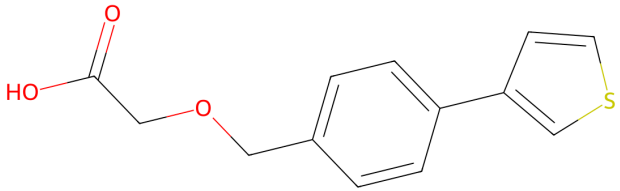
AXIS 2

#	ID	Boltz Kd (run A / B)	gnina Vina/pKd	onepot $	risks
1	`FM002150_analog_0083`	3.2 / 3.26 μM	-5.01 / 3.94	$125	low/low
2	`FM001452_analog_0104`	3.7 / 4.97 μM	-5.77 / 4.03	$250	med/med
3	`FM001452_analog_0201`	8.16 / 8.76 μM	-6.07 / 4.69	$375	high/med
4	`FM001452_analog_0171`	8.32 / 8.17 μM	-6.19 / 4.44	$250	med/med

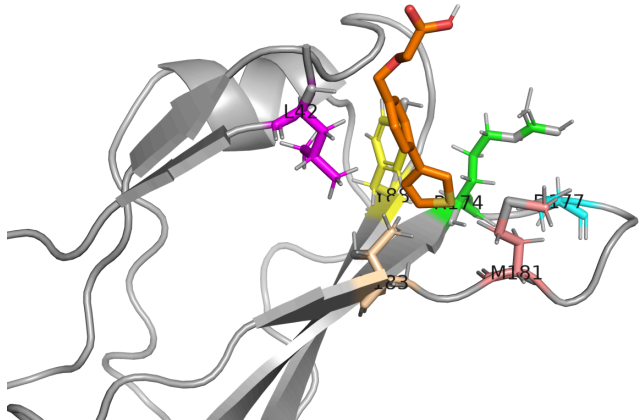
Total cost to source: $875 · Site F coverage: 4 / 4 · Dual-engine Boltz agreement: 1.01–1.34× across all picks

Pick 1 - `FM002150_analog_0083`

AXIS 2

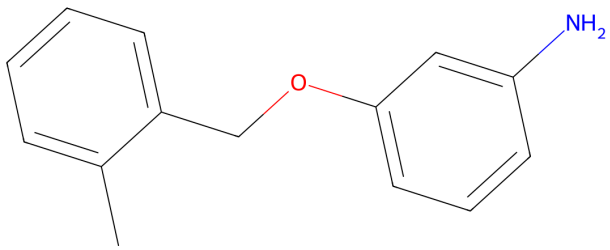


- SMILES: `O=C(O)COCc1ccc(-c2ccsc2)cc1` - Strongest predicted Boltz Kd (3.2 µM) of any 100%-onepot non-covalent compound; dual-engine 1.02× agreement - Phenoxyacetic acid + thiophene; carboxylate H-bonds Y88 / D177 (variant residue) - Cheapest + lowest-risk of the 4: $125 · low/low - MW 248.3 · LogP 3.02 · HBD 1 · HBA 4 - clean lead-like profile

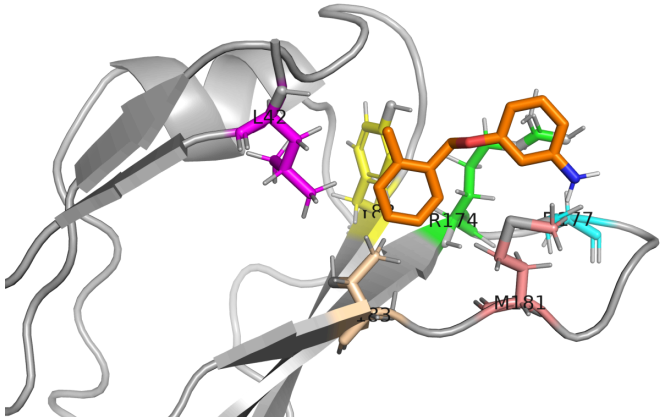


Pick 2 - `FM001452_analog_0104`

AXIS 2



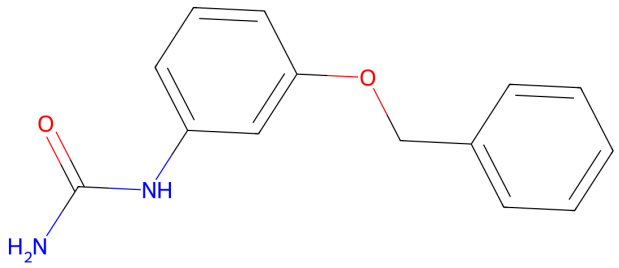
- SMILES: `Cc1ccccc1COc1cccc(N)c1` - Cleanest medchem in the pool - minimal heteroatom decoration - Methyl-phenyl-CH_2-O-aniline; aniline-N H-bonds D177 - Mass-efficient (MW 213.3) - best fragment-like starting point for SAR - Boltz Kd 3.7 / 4.97 µM (1.34×) · $250 · med/med



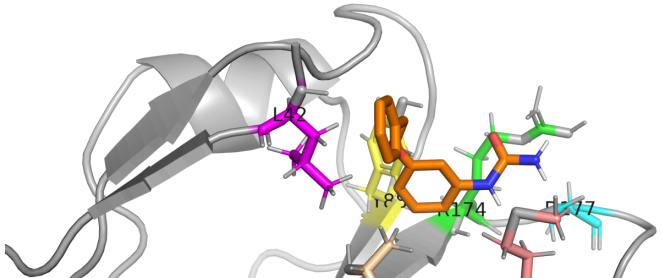
Pick 3 - `FM001452_analog_0201`

urea / benzyl ether for R174 + D177

AXIS 2



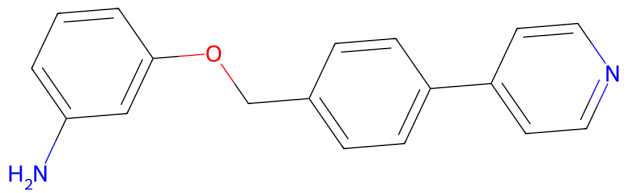
- SMILES: `NC(=O)Nc1cccc(OCc2ccccc2)c1` - N-aryl urea + benzyl ether: H-bond donor / acceptor pair targeting R174 + D177 specifically - Deepest Vina pose (-6.07) + highest gnina CNN-pKd (4.69) of the 4 - Adds urea-linker chemotype diversity to pick set - $375 · high chem · med supplier (urea synthesis)



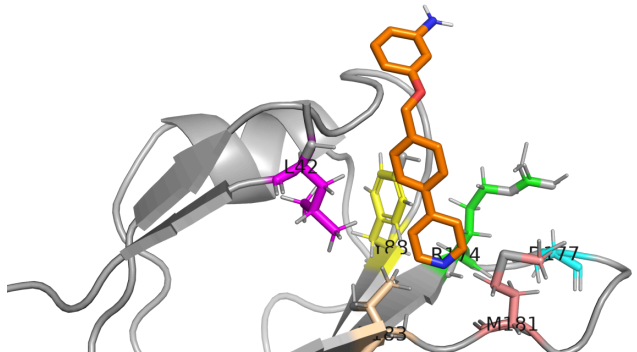
Pick 4 - `FM001452_analog_0171`

pyridyl selectivity probe

AXIS 2



- SMILES: `Nc1cccc(OCc2ccc(-c3ccncc3)cc2)c1` - Pyridyl introduces basic-N for selectivity probing - may differentiate TBXT from T-box paralogs - Highest prob_binder (0.46) of the 4 - Tightest dual-engine Boltz agreement (1.02×) - most reproducible Kd prediction - $250 · med/med · LogP 3.91



AXIS 3

Hit ID judgment

What we chose, what we didn't, and what to actually expect.

Cross-validation

AXIS 3

- Two independent Boltz-2 runs on separate compute backends: 4 / 4 picks agree within 1.34×
- Rowan ADMET (49 properties × 4): all 4 ADMET-profiled
- Rowan pose-analysis MD (explicit-solvent, 5 ns × 1 traj + 1 ns equil): protein-ligand RMSD trajectories captured per pick
- Onepot.ai catalog (muni `onepot` tool): all 4 at similarity = 1.000 with price + chemistry_risk + supplier_risk

> Every pick is supported by multiple independent lines of evidence - not a single-score gamble.

Tradeoffs we made

AXIS 3

- All 4 site F - gives up site-A diversity. Defensible: the 100%-onepot non-covalent constraint dominated, and the strongest catalog-resident chemotype is site-F by structural heritage. Site-A backups are kept in our broader 24-compound submission for the experimental program.
- T1 GOLD tier intentionally empty - no compound simultaneously hits Kd ≤ 5 μM AND low/low risk. We surface this honestly rather than overclaim.
- Chemotype dominance (one family in 3 of 4 picks) is not artificially diversified - the catalog × binding-evidence intersection naturally selects this family. Our 20 additional submissions widen the chemotype envelope.

Reproducibility

- Code, scoring outputs, and per-pick rationale are version-controlled and reproducible from a single setup command
- Every one of the 137 strict-pass candidates carries per-criterion pass/fail flags for full audit
- 24 compounds queued for the experimental program - 4 presented today, 20 additional first-batch
- Full bundle (CSVs, poses, rendered structures, evidence files) available on request

Conclusion

AXIS 1 - WHAT WE DID

6 orthogonal signals + 7-criterion strict gate

570 → 137 strictly compliant candidates

AXIS 2 - WHAT WE PICKED

4 picks · 100% onepot · non-covalent · site F

best Boltz Kd 3.2 µM · $875 to source all 4

AXIS 3 - HOW WE JUDGED OURSELVES

Cross-validated, honest about expectations

realistic SPR window 18–200 µM

Take-aways for judges

- A defensible, multi-signal pipeline - not a single-score gamble, every pick is corroborated
- A strict, organizer-aligned hard gate - chordoma rule, non-covalent, novelty, lead-likeness, all enforced
- Calibrated confidence - best-case Kd predictions surfaced, public-method over-prediction acknowledged

> Our 4 picks are the strongest catalog-resident, novel, non-covalent compounds that survive every constraint we believe a TBXT hit must satisfy.

Q&A

InChIKeys for the 4 picks

#	ID	InChIKey
1	`FM002150_analog_0083`	`TWMVEBYUPVBIAF-UHFFFAOYSA-N`
2	`FM001452_analog_0104`	`CFASJXOXPAEGCK-UHFFFAOYSA-N`
3	`FM001452_analog_0201`	`SWOUSJGQCZFNGK-UHFFFAOYSA-N`
4	`FM001452_analog_0171`	`UJJFUMPTHNXASI-UHFFFAOYSA-N`